Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam C · Form 1: Connecting ideas

Multiple Choice Questions: Select the best answer. (1 point each)

1. Two launches at equal speed use angles of 30° and 60° and land at their launch height. Which quantity is larger for the 60° launch?

(A) Horizontal speed

(B) Flight time

(C) Gravitational acceleration magnitude

(D) Range

2. A ball is launched at 15 m/s, 50° above horizontal, and lands at its launch height. Ignore drag. Find flight time. Use g = 9.8 m/s².

(A) -2.345 s

(B) 4.69 s

(C) 1.173 s

(D) 2.345 s

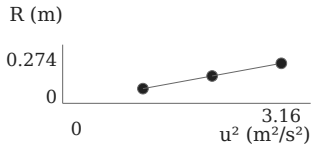
3. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u² (m²/s²) -> R (m): (1.16, 0.1005); (2.16, 0.1871); (3.16, 0.2737). Which interpretation of the experiment is valid?

(A) Horizontal interval alone determines vertical change without a model.

(B) The slope is the first vertical reading.

(C) Complementary angles give equal ranges but unequal flight times.

(D) Exchanging axes always preserves the numerical slope.



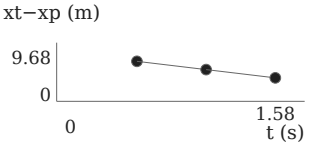
4. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.58, 9.68); (1.08, 7.68); (1.58, 5.68). By what factor does the first-to-third change exceed the first-to-second change?

(A) 2

(B) 2.8

(C) 3.6

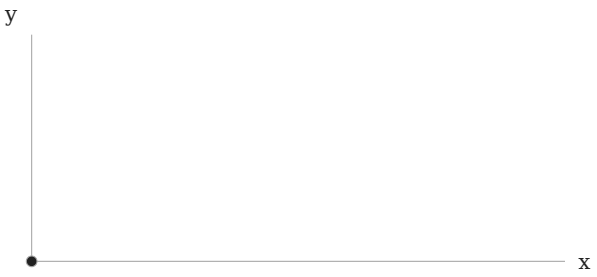
(D) 1.2



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball is launched from ground at 18 m/s, 30° above horizontal, and lands on the same ground. Ignore drag.

a. Draw the initial velocity vector and its horizontal and vertical components. **(1 pt)**



b. Find flight time, range, and velocity and acceleration at the top. **(3 pt)**

6. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. uy (m/s) -> tpeak (s): (1.16, 0.116); (2.16, 0.216); (3.16, 0.316).

a. Build the model for apex timing from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. A ball launched at speed u and angle θ lands at the same height without drag. Derive R = u²sin(2θ)/g from the component equations. **(3 pt)**